

## Project 2: Modeling Maximum Heart Rate Using Regression

**Assignment Type:** Individual

**Weighting:** This project encompasses the learning and application aspects of both the Homework and Lab components (collectively 20%).

**Learning Outcomes Assessed:** LO2, LO3

**Topic Coverage:** Weeks 7-10 (Data Visualization, Correlation, Linear Regression, Inference, ANOVA)

### Objective

To develop MATLAB computing skills for investigating relationships between health variables. You will apply correlation and linear regression techniques to model and predict the *Maximum Heart Rate Achieved* during exercise based on a person's *Age*.

### Problem Statement

You are working with a cardiology research team studying the effects of aging on physical fitness. They have provided you with a dataset (`Heart_Disease_Dataset.csv`) that contains the age of patients (*Age*) and their resulting maximum heart rate (*MaxHeartRate*) achieved during a stress test. Your goal is to model this relationship and evaluate its statistical significance.

### Dataset

The dataset `heart_rate_data.csv` contains two columns:

1. *Age*: The age of the patient in years.
2. *MaxHeartRate*: The maximum heart rate achieved (beats per minute).

## Tasks

### 1. Data Exploration and Visualization (LO2, LO3)

- Write a MATLAB script to import the data from the CSV file.
- Create a scatter plot of `MaxHeartRate` (y-axis) versus `Age` (x-axis). Label your axes correctly.
- Calculate the correlation coefficient ( $r$ ) between the two variables and interpret its value (discuss the strength and direction of the linear relationship).

### 2. Simple Linear Regression (LO2, LO3)

- Fit a simple linear regression model to the data, with `MaxHeartRate` as the response variable and `Age` as the predictor.
- Determine the estimated regression equation (i.e., find the intercept  $\beta_0$  and slope coefficient  $\beta_1$ ).
- Add the fitted regression line to your scatter plot from Task 1.

### 3. Inference and Model Diagnostics (LO2, LO3)

- Use MATLAB to find the 95% confidence intervals for the slope and intercept of the regression line.
- Test the hypothesis that the slope is zero (i.e.,  $H_0: \beta_1 = 0$ ) at a significance level of  $\alpha = 0.05$ . Report the p-value and state your conclusion.
- Calculate and plot the residuals. Create a residual plot (residuals vs. fitted values) to assess the assumptions of the linear regression model (linearity, constant variance).

### 4. Prediction and ANOVA Introduction (LO2, LO3)

- Use your fitted model to predict the expected maximum heart rate for two new patients: one who is **40 years old** and another who is **65 years old**.
- Perform a one-way Analysis of Variance (ANOVA) to test if there is a significant difference in the mean heart rate if the age data is categorized into three groups:
  - **"Young"** (29–45 years)
  - **"Middle-Aged"** (46–60 years)
  - **"Senior"** (61–77 years)

- Briefly explain the purpose of ANOVA in this context.

### **Deliverables**

- A single, well-commented MATLAB live script (Project2\_YourStudentID.mlx) that performs all the tasks.
- The script should display all numerical results (correlation coefficient, regression equation, confidence intervals, p-values, predictions) and all generated plots.
- A text box (using the MATLAB live script feature) at the end of the script that briefly summarizes your findings. This summary text box at the end answering: *"Is there a significant linear relationship between age and max heart rate? Based on your model, what is the expected decrease in maximum heart rate for every 10-year increase in age?"*

### **Generative AI Encouragement (as per Section 5.2)**

You are permitted to use GenAI tools like Microsoft Copilot to help you with this assignment.

- **How to use:** You may use it to help structure your code, to find the correct MATLAB function for performing an ANOVA test, or to ask for an explanation of the output from the regression analysis (e.g., "What does the p-value for the F-statistic in a linear regression tell me?").
- **Acknowledgment:** In the comments at the beginning of your MATLAB script, you **must** include a brief statement acknowledging which GenAI tool you used and for what purpose (e.g., "% Microsoft Copilot was used to find the 'anova1' function and to help interpret the ANOVA output table.").
- **Critical Thinking:** Remember to critically evaluate the GenAI's suggestions. For example, ensure the GenAI-recommended ANOVA test is the correct one for your categorized data and that you understand the underlying assumptions.